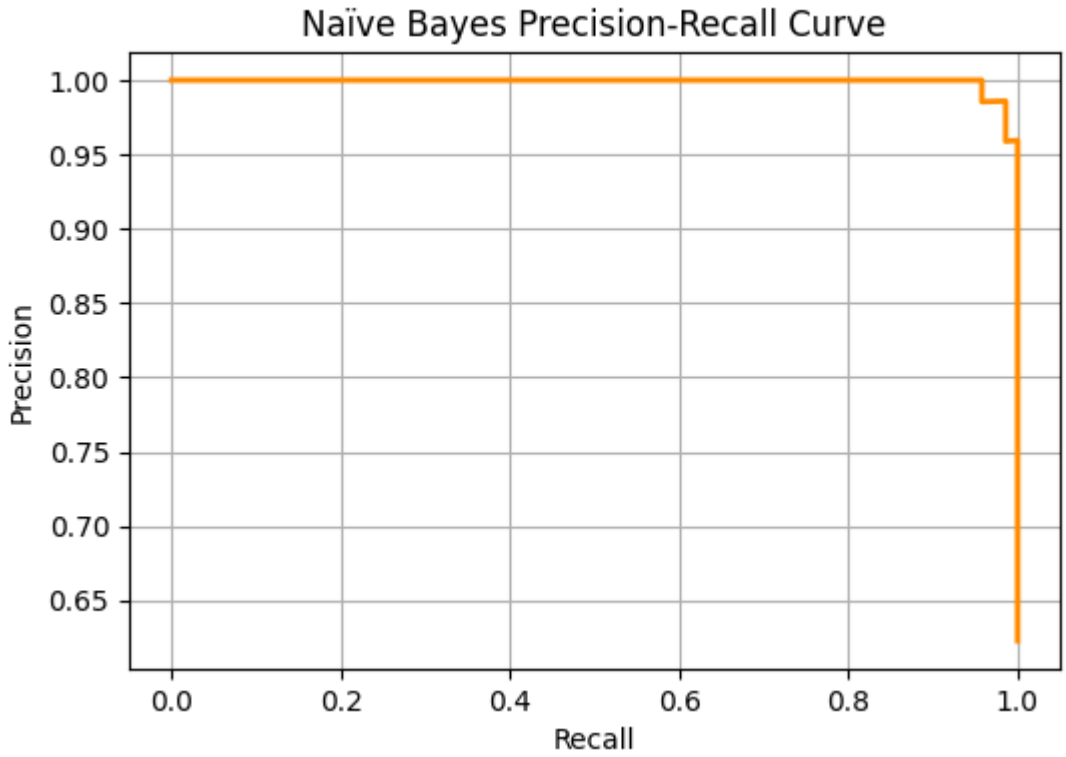
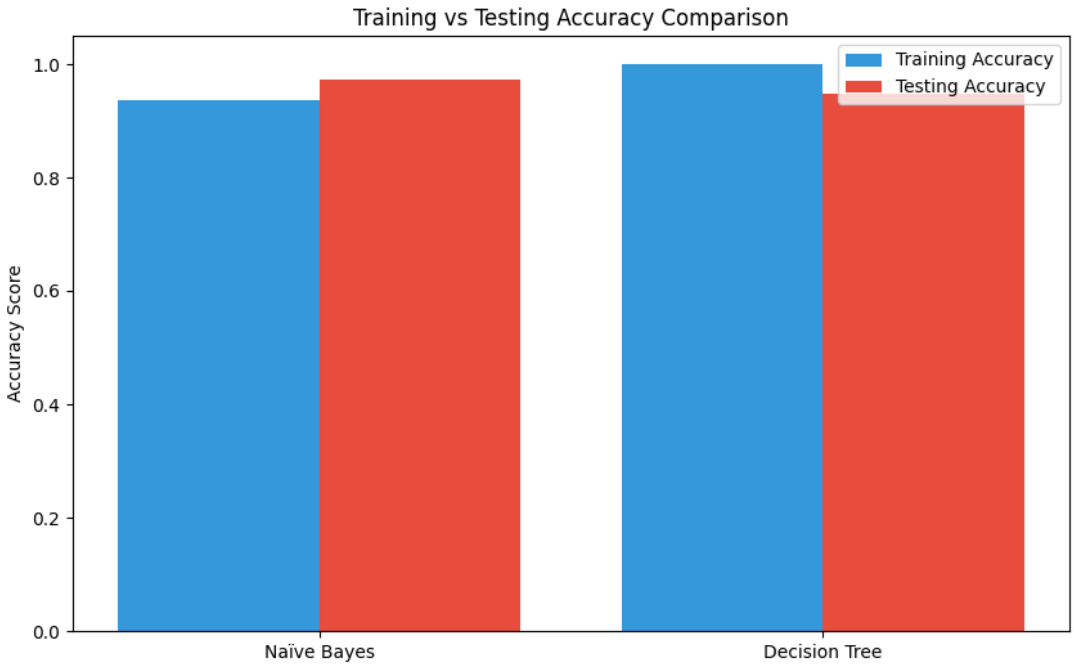
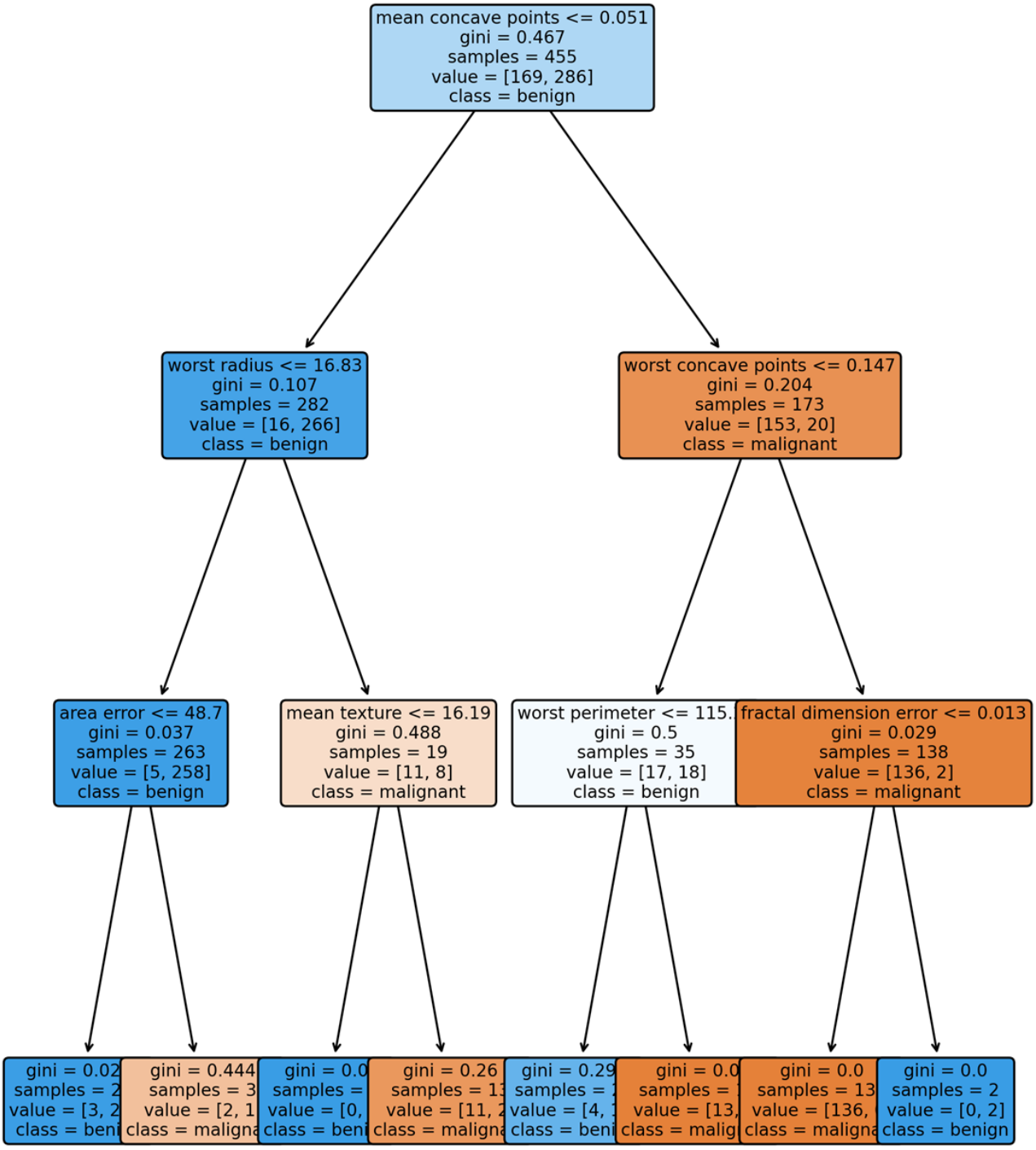
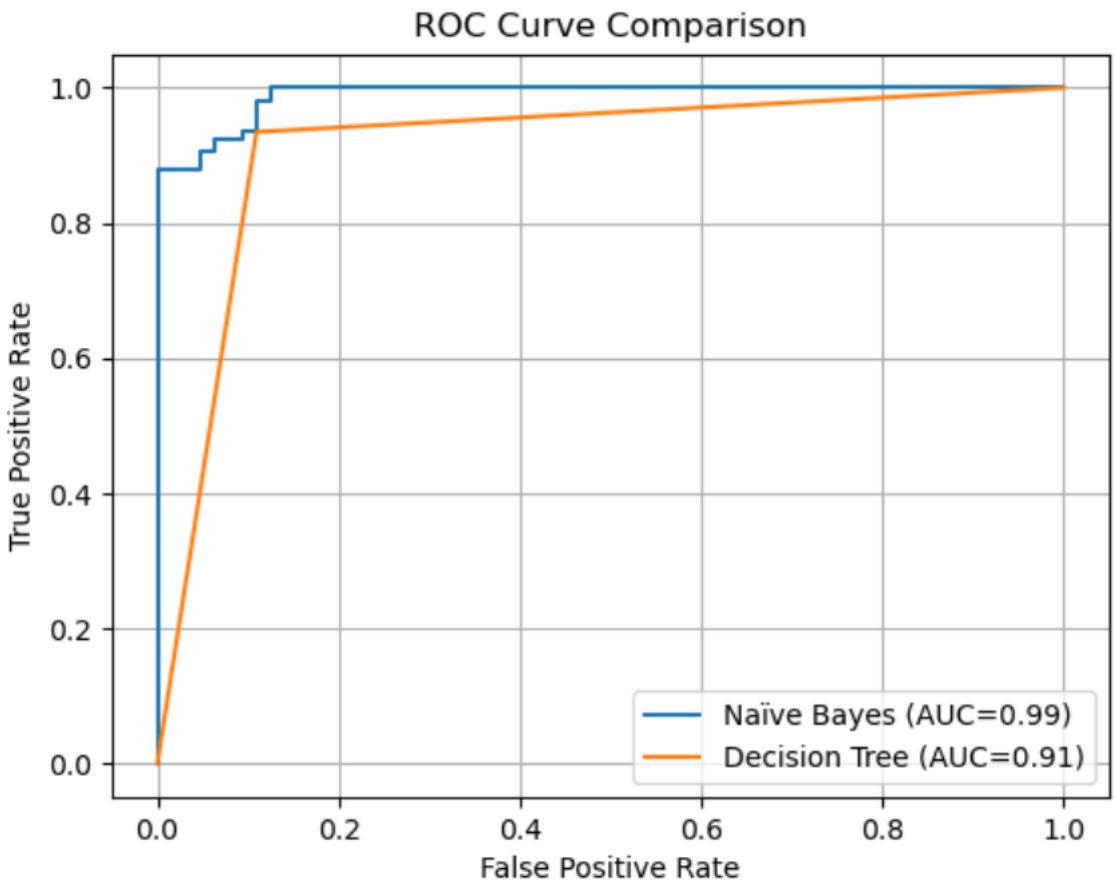

Naïve Bayes Precision-Recall Curve
Precision
Recall
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.0
0.2
0.4
0.6
0.8
1.0

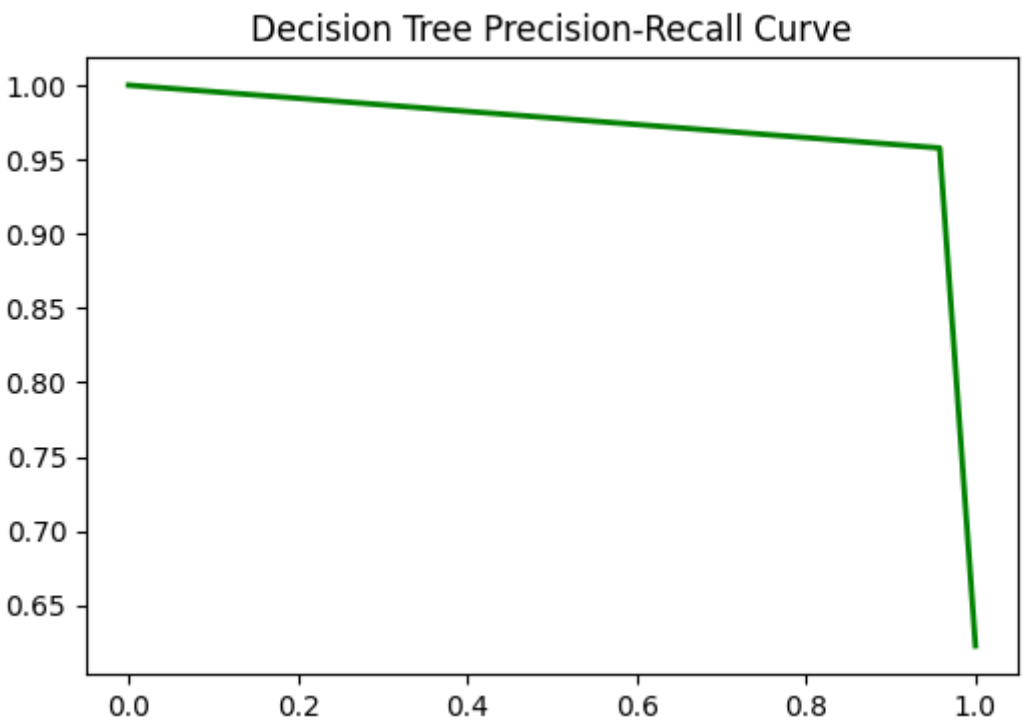

Training vs Testing Accuracy Comparison
Training Accuracy
Testing Accuracy
Accuracy Score
1.0
0.8
0.6
0.4
0.2
0.0
Naïve Bayes
Decision Tree

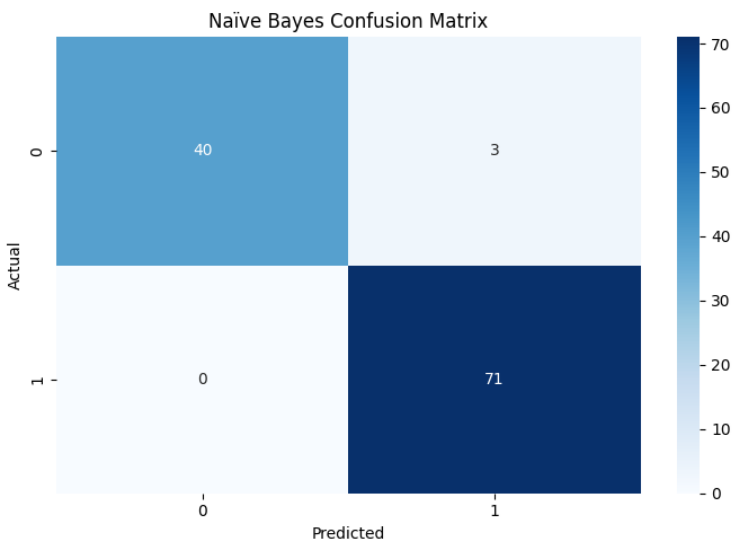
Decision Tree (Max Depth = 3)

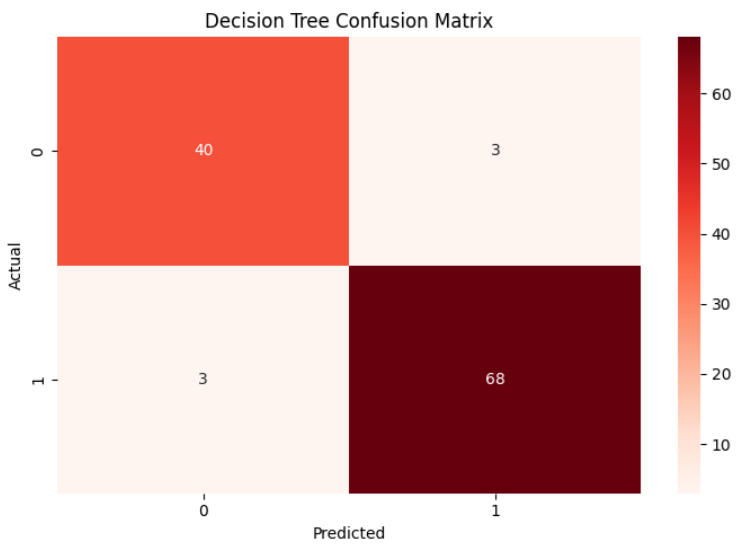


```
❯ python l5_e1.py; python3 l5_e2.py; python l5_e3.py
--- Experiment 1: Naïve Bayes Results ---
Accuracy: 0.9737

Confusion Matrix:
 [[40  3]
 [ 0 71]]

Classification Report:
              precision    recall  f1-score   support

           0       1.00      0.93      0.96        43
           1       0.96      1.00      0.98        71

    accuracy                           0.97       114
   macro avg       0.98      0.97      0.97       114
weighted avg       0.97      0.97      0.97       114

/home/abhinav/kaam/MLLab/l5_e1.py:34: UserWarning: FigureCanvasAgg is non-interactive, and
thus cannot be shown
  plt.show()
--- Experiment 2: Decision Tree Results ---
Accuracy: 0.9474

Confusion Matrix:
 [[40  3]
 [ 3 68]]
Tree visualization saved as ex2_decision_tree.png
PR Curve saved as ex2_pr_curve.png
Saved: ex3_accuracy_comparison.png
Saved: ex3_roc_comparison.png
Saved: ex3_cm_heatmaps.png
```




Decision Tree Precision-Recall Curve
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.0
0.2
0.4
0.6
0.8
1.0


Naïve Bayes Confusion Matrix
40
3
0
71
Actual
Predicted
0
1
70
60
50
40
30
20
10
0


Decision Tree Confusion Matrix
40
3
3
68
Actual
Predicted
0
1
60
50
40
30
20
10